

Task 1

To parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs

Code:

```
geya@LenovoL480: ~/OS/Skil × + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main()
{
    char filename[100];
    char text[200];

    printf("Enter input file name: ");
    scanf("%s", filename);

    int saved_stdin = dup(STDIN_FILENO);

    int fd = open(filename, O_RDONLY);

    if (fd == -1)
    {
        perror("Cannot open input file");
        return 1;
    }

    dup2(fd, STDIN_FILENO);
    close(fd);

    printf("Reading from redirected input:\n");
```

```

while (fgets(text, sizeof(text), stdin) != NULL)
{
    printf("%s", text);
}

dup2(saved_stdin, STDIN_FILENO);
close(saved_stdin);

printf("\nInput restored.\n");

return 0;
}

```

Output:

```

geya@LenovoL480:~/OS/Skill/Week 7$ ./task1
Enter input file name: input.txt
Reading from redirected input:

Hello from input file

Input restored.
geya@LenovoL480:~/OS/Skill/Week 7$ |

```

Task 2

To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection

Code:

```

#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main()
{
    char filename[100];

    printf("Enter output file name: ");
    scanf("%s", filename);

    int saved_stdout = dup(STDOUT_FILENO);

    int fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd == -1)
    {
        perror("Cannot open output file");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("This text was redirected to the output file.\n");
    printf("Output redirection is working.\n");

    dup2(saved_stdout, STDOUT_FILENO);
    close(saved_stdout);

    printf("Output restored.\n");

    return 0;
}

```

Output:

```

geya@LenovoL480:~/OS/Skill/Week 7$ ./task2
Enter output file name: output.txt
Output restored.
geya@LenovoL480:~/OS/Skill/Week 7$ cat output.txt
This text was redirected to the output file.
Output redirection is working.
geya@LenovoL480:~/OS/Skill/Week 7$ |

```

Task 3

To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main()
{
    char filename[100];

    printf("Enter file name to append: ");
    scanf("%s", filename);

    int saved_stdout = dup(STDOUT_FILENO);

    int fd = open(filename, O_WRONLY | O_CREAT | O_APPEND);

    if (fd == -1)
    {
        perror("Cannot open file");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("This line was appended to the file.\n");

    dup2(saved_stdout, STDOUT_FILENO);
    close(saved_stdout);

    printf("Append completed.\n");

    return 0;
}
```

Output:

```
geya@LenovoL480:~/OS/Skill/Week 7$ echo "Original data" > test.txt
geya@LenovoL480:~/OS/Skill/Week 7$ ./task3
Enter file name to append: test.txt
Append completed.
geya@LenovoL480:~/OS/Skill/Week 7$ cat test.txt
Original data
This line was appended to the file.
geya@LenovoL480:~/OS/Skill/Week 7$ |
```

Task 4

TO parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main()
{
    char filename[100];

    printf("Enter error output file name: ");
    scanf("%s", filename);

    int saved_stderr = dup(STDERR_FILENO);

    int fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd == -1)
    {
        perror("Cannot create error file");
        return 1;
    }

    dup2(fd, STDERR_FILENO);
    close(fd);

    fprintf(stderr, "This is an error message.\n");
    fprintf(stderr, "An error occurred during the operation.\n");
```

```
    dup2(saved_stderr, STDERR_FILENO);  
    close(saved_stderr);  
  
    printf("stderr restored.\n");  
  
    return 0;  
}
```

Output:

```
geya@LenovoL480:~/OS/Skill/Week 7$ ./task4  
Enter error output file name: error.txt  
stderr restored.  
geya@LenovoL480:~/OS/Skill/Week 7$ cat error.txt  
This is an error message.  
An error occurred during the operation.  
geya@LenovoL480:~/OS/Skill/Week 7$ |
```